WHAT IS A SOC?

by Lorenzo Levano

Lorenzo Levano

What is a SOC in IT Companies?

A Security Operations Center (SOC) in IT companies is like a high-tech control room where a team of experts works to protect the company's computers, data, and networks from intruders.

Why Do IT Companies Need a SOC?

IT companies deal with a lot of important and sensitive information, such as customer data, business secrets, and financial details. If this information falls into the wrong hands, it can cause serious problems.

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WHAT DOED A SOC DO?

Watches Over the Company

The SOC keeps an eye on the company's computers, networks, and other digital assets 24/7 thanks to special software that detect unusual activities.

Stops Threats in Their Tracks

If something suspicious is found, the SOC team acts quickly to stop it before it causes any harm.

Investigates Problems

When something goes wrong, the SOC investigates to find out what happened, how it happened, and what needs to be done to prevent it from happening again.

Keeps the Company Safe

The SOC doesn't just react to problems—they also work proactively to strengthen the company's defenses.

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HOW DOES A SOC WORK?

01

People

The SOC is staffed by a team of cybersecurity experts who know how to find and stop cyber threats.

02

Processes

The SOC follows a set of rules and procedures to handle any security incident.

03

Technology

The SOC uses advanced tools and software to monitor the company's digital environment.

ROLES IN CYBERSECURITY

by Lorenzo Levano

Lorenzo Levano

SECURITY
ANALYST

- **What They Do:** Security Analysts are like detectives who look for signs of cyber attacks. They monitor the company's computers and networks to spot anything suspicious.
- **Main Tasks:** Checking security alerts, investigating potential threats, and helping to fix any security issues they find.
- **Example:** If an alarm goes off in a building, the Security Analyst checks to see if there's a real problem and takes steps to fix it.

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SECURITY ENGINEER

- What They Do: Security Engineers build and maintain the company's security systems. They design and set up tools and processes to protect the company from cyber threats.
- Main Tasks: Setting up firewalls, encryption, and other security measures to protect data and systems.
- Example: If you want to build a secure vault for your valuable items, a Security Engineer designs and builds the vault and makes sure it's strong and secure.

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SECURITY
ARCHITECT

- What They Do: Security Architects design the overall security strategy and framework for a company. They create plans to ensure that all systems and data are protected effectively.

- Main Tasks: Developing security policies and ensuring that all security measures work together smoothly.

- Example: If you're designing a secure fortress, a Security Architect decides how the fortress should be built, what kind of walls and defenses to use, and how to keep it secure.

Lorenzo Levano

INCIDENT RESPONDER

- What They Do: Incident Responders are like emergency responders who handle cyber attacks when they occur. They act quickly to fix problems and minimize damage.
- Main Tasks: Identifying and managing security incidents, recovering affected systems, and preventing future incidents.
- Example: If a break-in occurs at the fortress, the Incident Responder deals with the situation, repairs any damage, and makes sure it doesn't happen again.

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PENETRATION TESTER

- What They Do: Penetration Testers, also known as Ethical Hackers, try to break into systems to find weaknesses before real hackers can exploit them.

- Main Tasks: Testing systems for vulnerabilities, finding security flaws, and recommending fixes to improve security.

- Example: If you're trying to test the strength of a vault, a Penetration Tester tries to crack it open to find out if it's really secure.

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SECURITY CONSULTANT

- What They Do: Security Consultants provide expert advice on how to improve a company's security. They assess current security measures and suggest improvements.
- Main Tasks: Reviewing security practices, providing recommendations, and helping with security planning and strategy.
- Example: If you need advice on how to secure your home, a Security Consultant looks at what you have and suggests ways to make it safer.

CISO

CHIEF INFORMATION SECURITY OFFICER

- What They Do: The CISO is the top person in charge of all cybersecurity efforts in a company. They oversee the entire security team.
- Main Tasks: Setting the overall security strategy and ensuring that the company's security policies are followed.
- Example: The CISO makes important decisions about how to protect and lead the security team.

SOC MANAGER

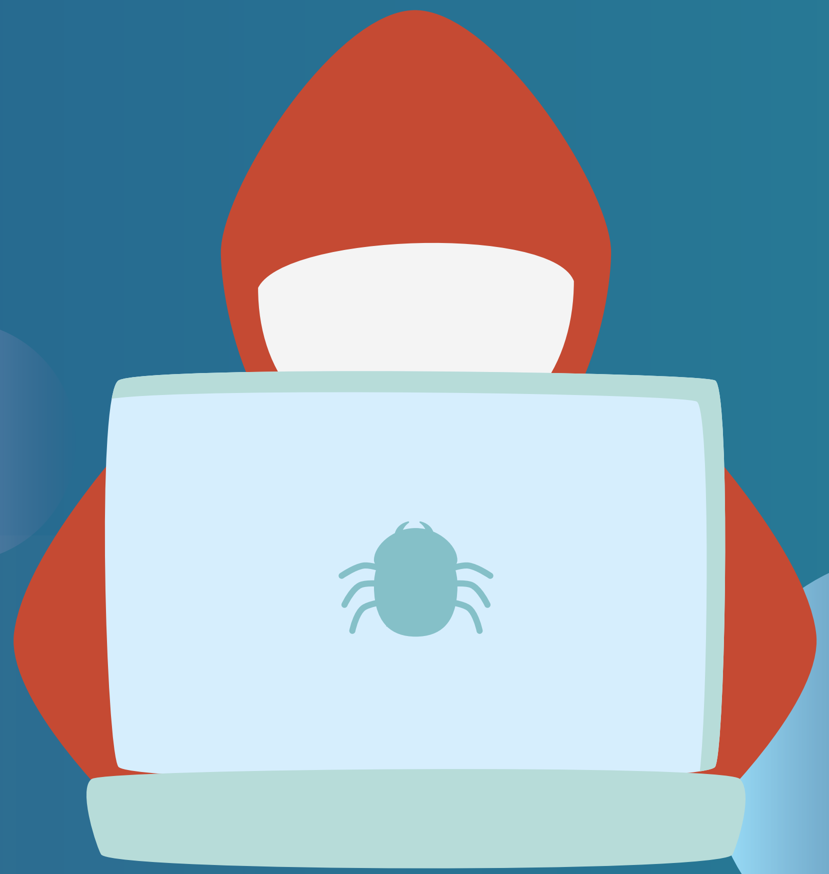
- What They Do: SOC Managers oversee the day-to-day operations of the SOC. They make sure that the SOC team is working effectively.
- Main Tasks: Managing SOC staff, coordinating incident responses, and ensuring that security monitoring tools are used correctly.
- Example: The SOC Manager is like the supervisor of a security control room, ensuring that everything runs smoothly and that any security problems are addressed quickly.

SOC'S STRUCTURE

by Lorenzo Levano

Lorenzo Levano

RED TEAM



- What They Do: Their job is to simulate real-world cyber attacks to find weaknesses in the company's security.
- Purpose: They test how well the company's defenses hold up against actual hacking attempts. This helps identify vulnerabilities that need to be fixed before real hackers can exploit them.
- Example: If the company's security system is a fortress, the Red Team tries to break in, just like a real thief would.

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BLUE TEAM

- What They Do: They monitor for any signs of attacks and respond to any security incidents.
- Purpose: Their goal is to protect the company from real attacks and to respond quickly if an attack happens.
- Example: If the Red Team tries to break into the fortress, the Blue Team is the security team inside the fortress, making sure it stays secure and responding to any intrusions.

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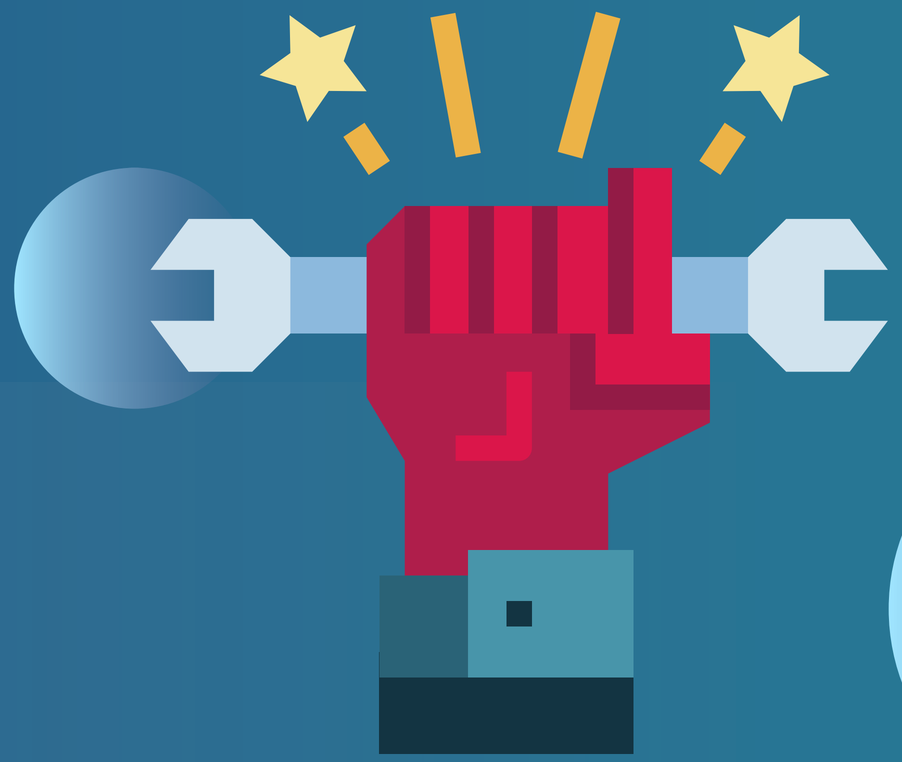
PURPLE TEAM

- What They Do: The Purple Team is a combination of the Red and Blue Teams. They improve the overall security of the company by sharing information and coordinating their efforts.
- Purpose: They help bridge the gap between the attackers (Red Team) and defenders (Blue Team).
- Example: The Purple Team acts like a special unit that helps both the attackers and defenders understand each other's methods, improving how the fortress is protected.



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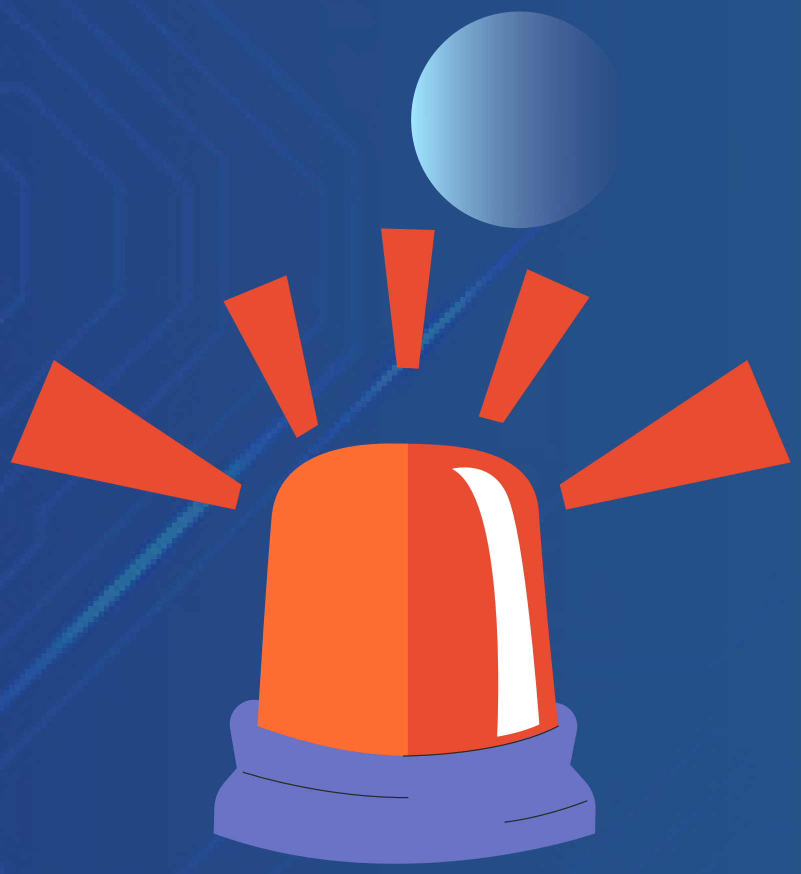

INCIDENT RESPONSE
TEAM

- What They Do: They investigate what happened, contain the problem, and work to fix it.
- Purpose: Their job is to quickly respond to any security breaches, minimize damage, and ensure that similar incidents don't happen again.
- Example: If the fortress is breached, the Incident Response Team is like the emergency responders who quickly deal with the situation and repair any damage.

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THREAT INTELLIGENCE TEAM

- What They Do: They track hacker tactics, vulnerabilities, and new types of malware.
- Purpose: Their goal is to provide the other teams with valuable information that helps them anticipate and prepare for potential threats.
- Example: The Threat Intelligence Team acts like scouts who gather information about potential threats and share it with the rest of the security teams to keep the fortress safe.

TYPES OF HACKERS

by Lorenzo Levano

Lorenzo Levano

WHITE HAT HACKERS

- What They Do: White Hat Hackers are the good guys. They use their skills to help organizations improve their security.
- Intentions: They are hired by companies to find and fix security vulnerabilities before malicious hackers can exploit them.
- Example: The hacker will try to find vulnerabilities and then report them so the company can fix them.

Activities:

- Performing security testing and audits to find weaknesses.
- Providing recommendations to strengthen security.
- Following ethical guidelines and working with permission.

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GREY HAT HACKERS

- What They Do: They often look for security flaws without being asked but do not use their findings for malicious purposes.
- Intentions: They may not have permission to test a system but usually report vulnerabilities to the organization.
- Example: A Grey Hat Hacker might find a vulnerability in a company's website and notify the company about it, but they didn't have permission to test the site in the first place.

Activities:

- Finding and sometimes disclosing security flaws without permission.
- Sometimes asking for a reward or payment to fix the issue.
- Their actions can be legal or illegal, depending on how they handle the information.

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BLACK HAT HACKERS

- What They Do: Black Hat Hackers are the bad guys. They use their skills to exploit vulnerabilities for personal gain or to cause harm.
- Intentions: They break into systems to steal data, spread malware, or cause damage without permission or ethical considerations.
- Example: A Black Hat Hacker might break into a company's database to steal customer information and sell it on the dark web.

Activities:

- Conducting illegal activities such as stealing personal information, installing ransomware, or damaging systems.
- Selling stolen data or using it for fraudulent purposes.
- Operating outside the law and ethical boundaries.

CIA TRIAD

by Lorenzo Levano

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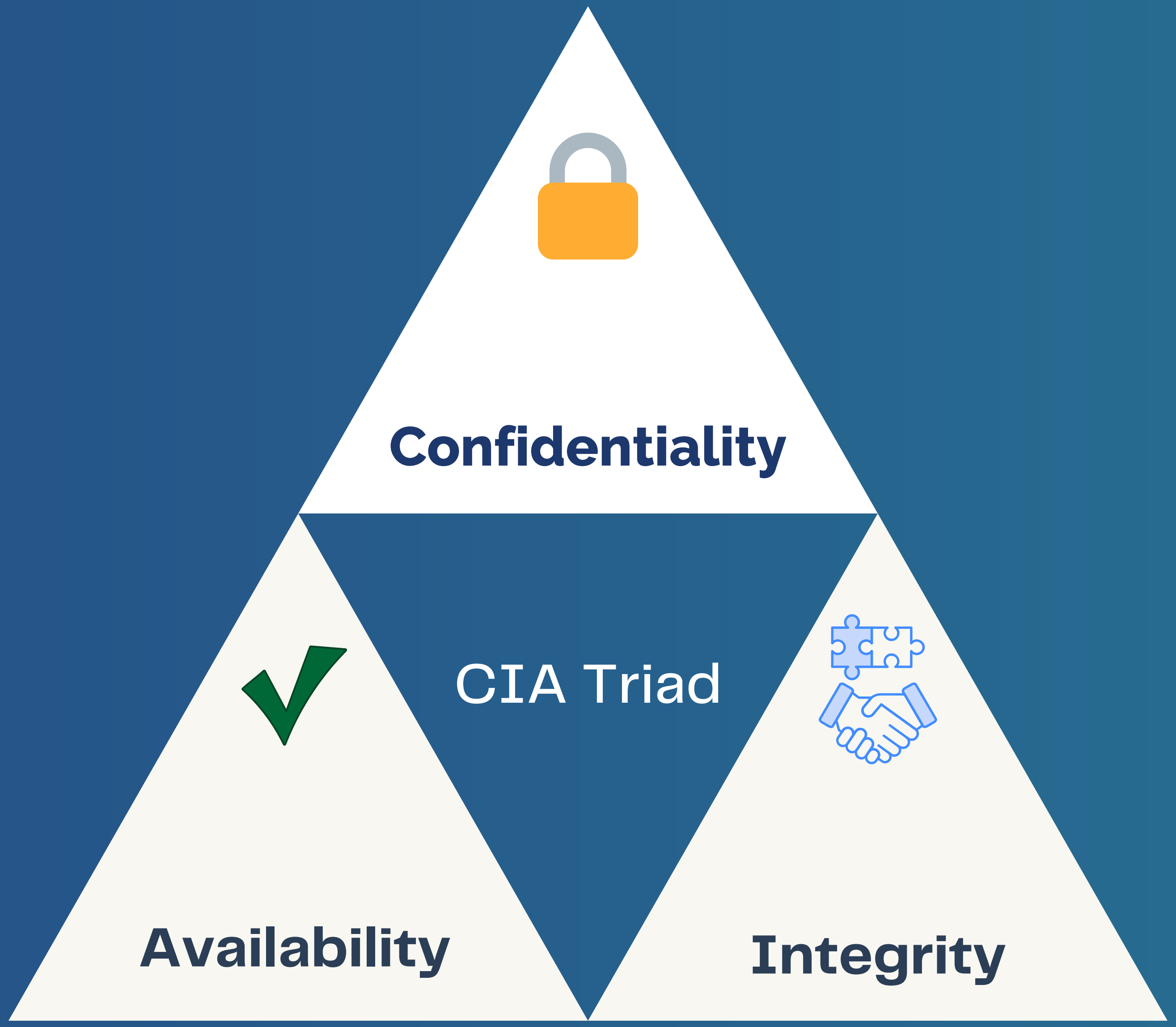

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CONFIDENTIALITY

- What It Means: Confidentiality means keeping information private and secure, so only authorized people can access it.
- Purpose: To prevent unauthorized individuals from seeing or using sensitive information.
- Example: Imagine a diary with a lock on it. Only people with the key can read what's inside. That's like confidentiality for your information.

How It Works:

- Using passwords to protect accounts.
- Encrypting data so that even if someone intercepts it, they can't read it without the key.

INTEGRITY

- What It Means: Integrity means ensuring that information is accurate and has not been tampered with or altered by unauthorized people.
- Purpose: To make sure that data remains correct and trustworthy.
- Example: Think of a sealed envelope with your important documents. If the seal is intact, you know that nobody has opened it and altered the contents. That's like integrity for your information.

How It Works:

- Using checksums or hashes to verify that data hasn't been changed.
- Implementing version control to keep track of changes to documents.

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AVAILABILITY

- What It Means: Availability means making sure that information and resources are accessible to authorized users when they need them.
- Purpose: To ensure that systems and data are up and running and can be used as needed.
- Example: Imagine a library that is always open so that you can access the books whenever you need them, ensuring availability.

How It Works:

- Using backup systems to recover data if it is lost.
- Implementing redundancies and load balancing to keep systems running smoothly.

ZERO TRUST

by Lorenzo Levano

Lorenzo Levano

No Assumptions

Zero Trust doesn't trust anyone or anything by default, even if they're inside the network.

Continuous Verification

Every time someone or something tries to access a system, data, or application, Zero Trust checks if they should be allowed.

Limit Access

Even if you're verified, you only get access to the specific resources you need.

Assume Breach

Zero Trust operates under the idea that a breach could happen at any time.

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HOW ZERO TRUST WORKS

01

User Verification

Every time you try to log in or access something, the system checks your identity using methods.

02

Device Verification

The system also checks if the device you're using is secure and allowed to access the network.

03

Limiting Access

Even after verifying, you can only access what's necessary for your job, not everything in the network.

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WHY IT'S IMPORTANT

Stronger Security

By verifying every request and limiting access, Zero Trust makes it much harder for attackers to steal data or cause harm.

Adapts to Modern Threats

With more people working remotely and using various devices, Zero Trust helps ensure security even when people aren't in the office.

WHAT IS A SIEM?

by Lorenzo Levano

Lorenzo Levano

DATA COLLECTION

- What It Means: SIEM collects data from various sources within a company's network, like servers, firewalls, and applications. This data includes logs, which are records of what's happening on these systems.
- Example: If someone logs into a company's server, the SIEM system will collect a log of that event.

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INTEGRITY
ANALYSIS

- What It Means: Once the data is collected, SIEM analyzes it to look for anything unusual or suspicious that might indicate a security threat.
- Example: If the SIEM notices that someone is trying to log in repeatedly and failing, it might flag this as a potential security threat, like someone trying to guess a password.

Lorenzo Levano

ALERTING

- What It Means: If the SIEM system finds something suspicious, it sends an alert to the security team so they can investigate further.
- Example: If the SIEM detects a strange login from a country where the company doesn't operate, it will alert the security team to check if it's a legitimate activity or a potential attack.

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REPORTING

- What It Means: SIEM also creates reports that show the overall security status of the company, helping the team see trends, identify potential weak spots, and improve security over time.
- Example: A SIEM report might show that there has been an increase in failed login attempts over the past month, suggesting that the company should strengthen its password policies.

Lorenzo Levano

WHY SIEM IS IMPORTANT

01

Centralized Monitoring

SIEM gathers all security data in one place, making it easier for the security team to monitor everything at once.

02

Faster Detection

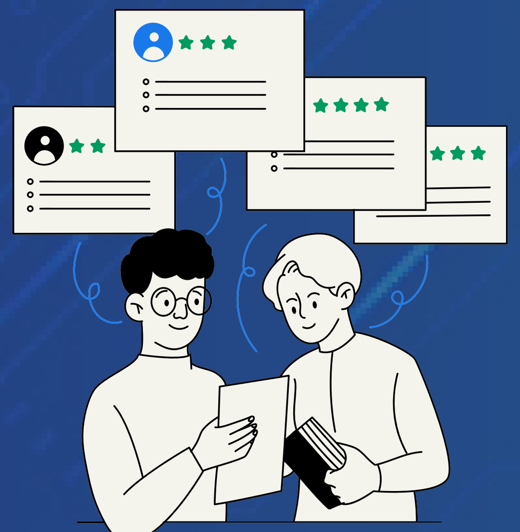
By analyzing data in real-time, SIEM helps detect potential security threats quickly.

03

Improved Compliance

SIEM helps companies meet regulatory requirements by providing detailed records and reports of security activities.

EXAMPLE OF SIEM



Collecting Data

Analyzing Data

Sending Alerts

WHAT IS A SOAR?

by Lorenzo Levano

Lorenzo Levano

SECURITY ORCHESTRATION

- What It Means: Orchestration is like conducting an orchestra, where different tools and systems work together in harmony. In cybersecurity, orchestration means connecting various security tools and systems so they can work together automatically.

- Example: If a security alert is detected by one tool, orchestration ensures that this alert is shared with other tools and systems without needing a person to do it manually.

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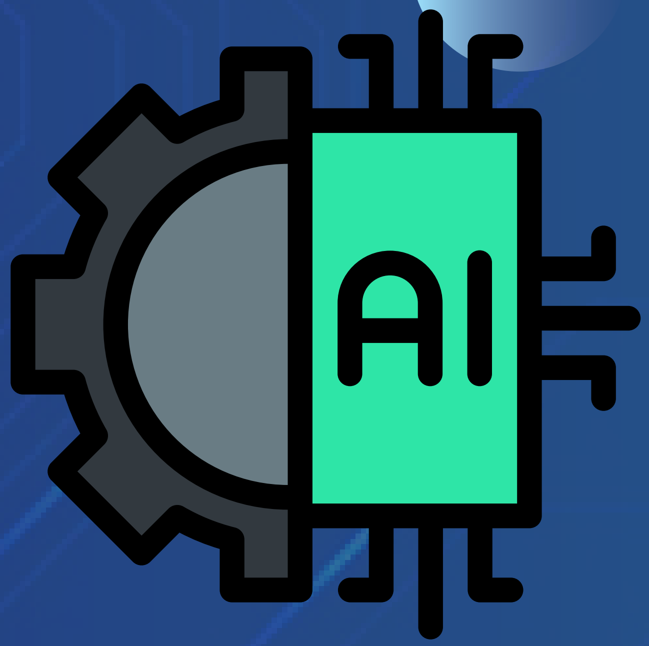
AUTOMATION



- What It Means: Automation in SOAR refers to setting up automatic responses to certain security events. Instead of waiting for a person to take action, the system can handle routine tasks by itself.
- Example: If a suspicious email is detected, the system can automatically quarantine it or block it from reaching users, without needing a human to intervene.

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RESPONSE

- What It Means: Response is about taking action when a security threat is detected. SOAR helps by providing quick and consistent responses to threats, which can be either fully automated or assisted by human analysts.

- Example: If an attack is detected on the network, SOAR can automatically isolate the affected parts of the network to prevent further damage.

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WHY SOAR IS IMPORTANT

01

Saves Time

By automating routine tasks, SOAR frees up cybersecurity teams to focus on more complex issues that need human judgment.

02

Improves Accuracy

Automation reduces the chance of human error, ensuring that responses to threats are consistent and fast.

03

Integrates Tools

SOAR brings together different security tools into one system, making it easier for teams to manage and respond to threats.

EXAMPLE OF SOAR

WHAT IS A FIREWALL?

by Lorenzo Levano

Lorenzo Levano

MONITORING TRAFFIC

- **What It Means**: A firewall watches the data (also called "traffic") that tries to enter or leave your computer or network. This data could be things like emails, files, or web pages.

- **Example**: Imagine your computer is like a house, and the firewall is a security guard at the front door. The guard checks everyone who tries to come in or out.

Lorenzo Levano

ALLOWING OR BLOCKING

DATA

- **What It Means**: The firewall decides if the data is safe or dangerous. If it's safe, the firewall lets it through. If it's dangerous, the firewall blocks it to keep your computer or network safe.
- **Example**: If someone tries to send a virus to your computer, the firewall blocks it, just like a security guard stopping a suspicious person from entering the house.

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SETTING
RULES

- **What It Means**: Firewalls operate based on rules set by the user or organization. These rules tell the firewall what kind of data is allowed and what should be blocked.

- **Example**: You might set a rule that blocks all data coming from a certain website you don't trust, similar to a rule that the security guard shouldn't let anyone in from a particular address.

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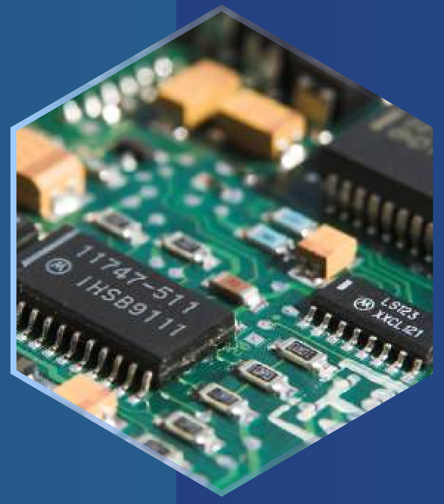
TYPES OF FIREWALLS

Software Firewall

- What It Is: A program installed on your computer that monitors and controls traffic.
- Example: Many operating systems, like Windows or macOS, come with built-in software firewalls.



Hardware Firewall

- What It Is: A physical device that sits between your network and the internet, monitoring traffic before it even reaches your computer.
- Example: A router with firewall features that protects all devices in your home network.

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WHY FIREWALL IS IMPORTANT

01

Protects Against Threats

Firewalls help keep out hackers, viruses, and other malicious software that could harm your computer or steal your information.

02

Controls Access

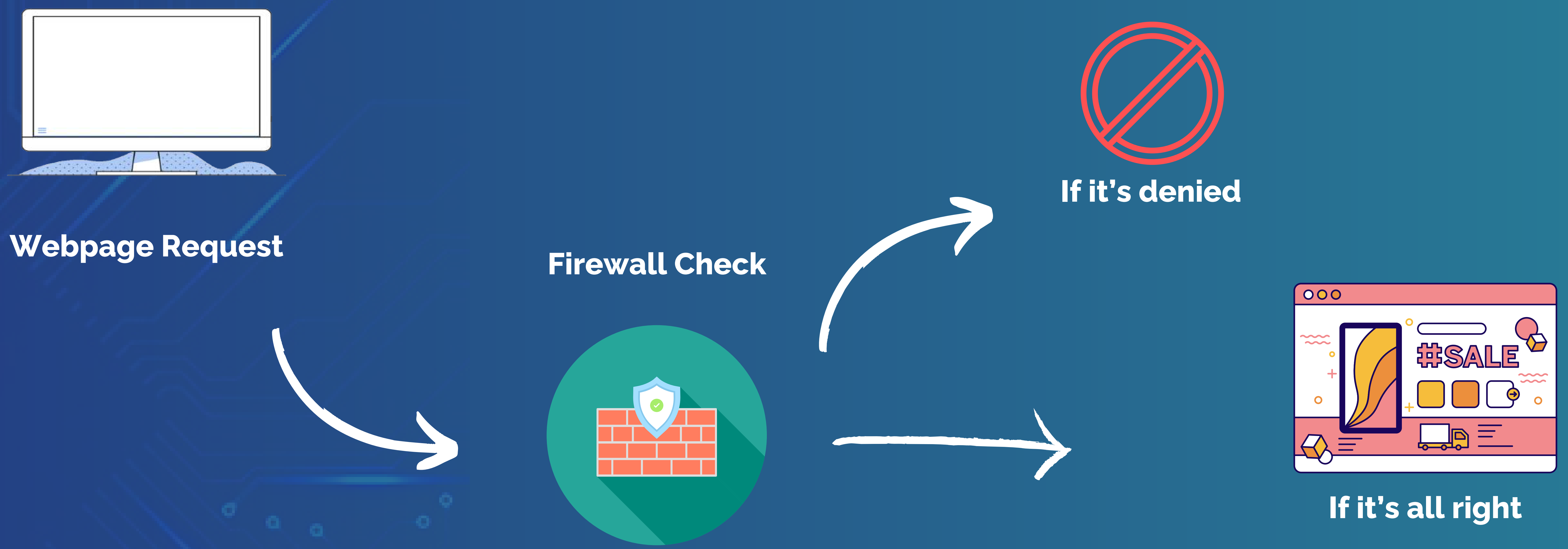
They can block access to dangerous websites or prevent unauthorized people from accessing your network.

03

Monitors Traffic

Firewalls keep an eye on what's happening on your network, providing logs and reports that help you understand if someone is trying to attack your system.

EXAMPLE OF FIREWALL



WHAT IS AN EDR?

by Lorenzo Levano

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MONITORING
ENDPOINTS

- **What It Means**: EDR constantly watches over your devices to see what's happening on them. It looks for anything unusual or suspicious, like strange programs running or files being accessed in unusual ways.
- **Example:** Imagine you have a security camera in your house that watches everything you do. If someone tries to break in, the camera catches it on video.

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DETECTING
THREATS

- **What It Means**: If EDR notices something that doesn't seem right—like a virus trying to install itself, or someone trying to hack into your computer—it quickly identifies this as a potential threat.
- **Example**: The security camera notices a person trying to force open your front door and recognizes this as a break-in attempt.

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RESPONDING
TO THREATS

What It Means: EDR doesn't just detect threats; it can also respond to them. Depending on the situation, it might:

- Alert you so you can take action.
- Block the threat automatically to prevent damage.
- Remove harmful software if it's already on the device.

Example: The security camera not only records the break-in attempt but also triggers an alarm to scare off the intruder.

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COLLECTING
DATA

- **What It Means**: EDR keeps detailed records of everything that happens on the device. This helps security teams understand what happened, how it happened, and what needs to be done to prevent it in the future.
- **Example**: After the break-in attempt, the security camera provides video footage that shows exactly what the intruder did, helping you prevent future break-ins.

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WHY EDR IS IMPORTANT

01

Protects Devices

EDR focuses on securing individual devices from threats, ensuring that your endpoint stays safe from hackers, viruses, and malware.

02

Quick Response

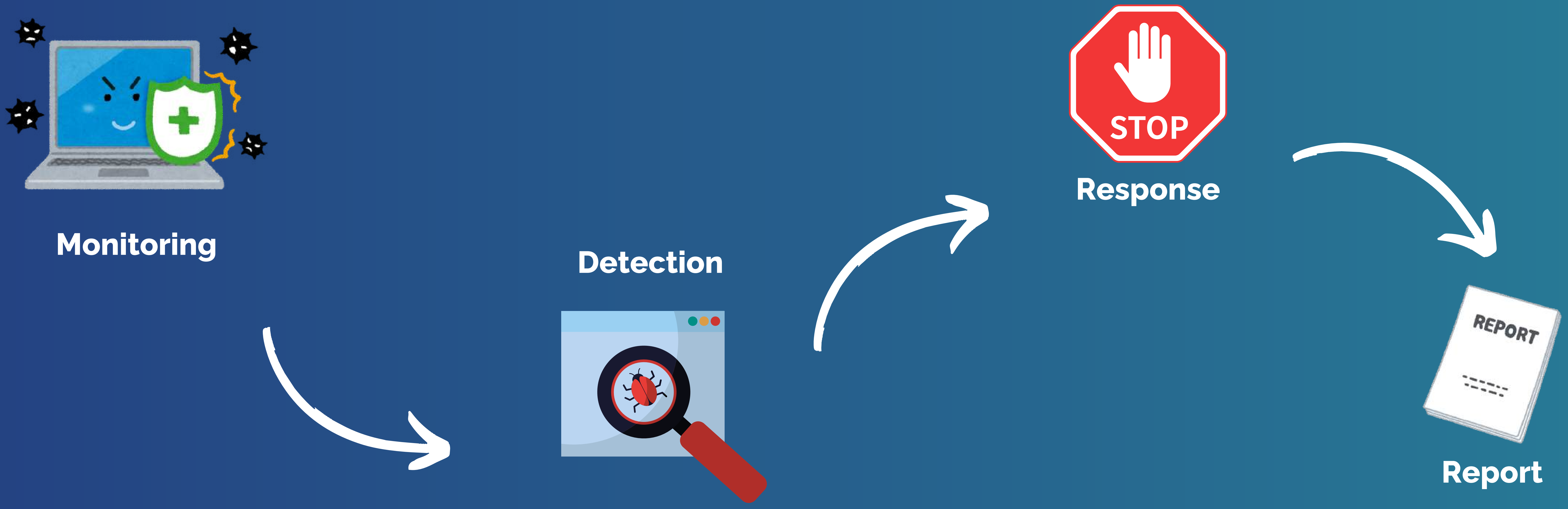
By detecting and responding to threats in real-time, EDR helps stop attacks before they can do serious damage.

03

Detailed Insights

EDR provides valuable information about security incidents, helping organizations improve their defenses over time.

EXAMPLE OF EDR



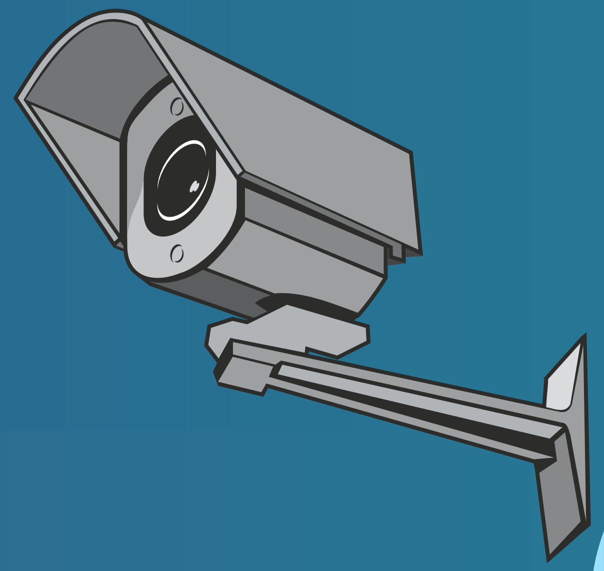
WHAT ARE IDS & IPS?

by Lorenzo Levano

Lorenzo Levano

IDS

WHAT IDS DOES



- IDS is like a security camera that watches over your network. It monitors all the data (traffic) coming in and going out of your network, looking for anything suspicious.

- **Detection Only:** IDS doesn't stop anything by itself; it just alerts you if it sees something unusual or potentially harmful.

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IDS

HOW IDS WORKS

- **Monitoring**: IDS continuously checks the network for signs of threats, like hackers trying to break in or malware trying to spread.
- **Alerting**: When IDS detects something that looks suspicious, it sends an alert to the security team. The team then investigates to see if it's a real threat.

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EXAMPLE OF IDS

Imagine you have a security camera in your house. If it sees someone sneaking around outside, it doesn't stop them, but it sends you an alert so you can take action.

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IPS

WHAT IPS DOES

- IPS is like a security guard that not only watches your network but also takes action to stop threats.
- **Detection and Prevention**: IPS monitors the traffic like IDS, but it also has the power to block or prevent the harmful data from getting into your network.

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HOW
IPS WORKS

- **Monitoring:** Just like IDS, IPS looks for suspicious activity in the network.
- **Blocking**: If IPS finds something dangerous, like a hacker trying to get in, it can automatically block the traffic or stop the attack right away.

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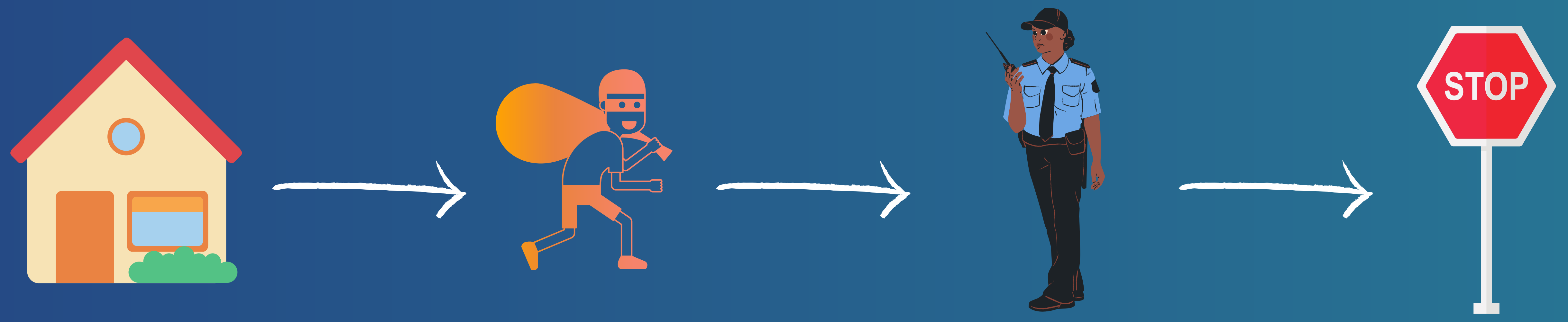
EXAMPLE OF IPS



Imagine you have a security guard at your front door. If someone tries to break in, the guard doesn't just alert you; they stop the intruder from entering.

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DIFFERENCES

BETWEEN IDS & IPS

IDS

- Watches and Alerts: It detects threats and sends alerts but doesn’t stop them.
- Response Needed: The security team needs to take action after an alert.

IPS

- Watches and Blocks: It detects threats and can automatically block them from entering the network.
- Automatic Response: It takes immediate action to prevent the threat, without waiting for the security team.

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WHY THEY ARE IMPORTANT

IDS

IDS helps by providing early warnings about potential threats, giving the security team time to respond.

IPS

IPS goes a step further by actively blocking threats, providing stronger and faster protection.

RECAP OF SOC
SOFTWARES

by Lorenzo Levano

Lorenzo Levano

SIEM
(SECURITY INFORMATION AND EVENT MANAGEMENT)

- **What It Does**: SIEM is like a central monitoring system. It collects data from all over the network (like logs and alerts), analyzes it to spot anything suspicious, and alerts the security team if something looks wrong.
- **Key Function**: Gathers and analyzes data to help detect potential security threats.

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SOAR

SECURITY ORCHESTRATION, AUTOMATION, AND RESPONSE

- **What It Does**: SOAR is like an automated assistant for the SOC. It helps different security tools work together, automates routine tasks (like responding to alerts), and ensures quick action when a threat is detected.

- **Key Function**: Automates and coordinates responses to security incidents.

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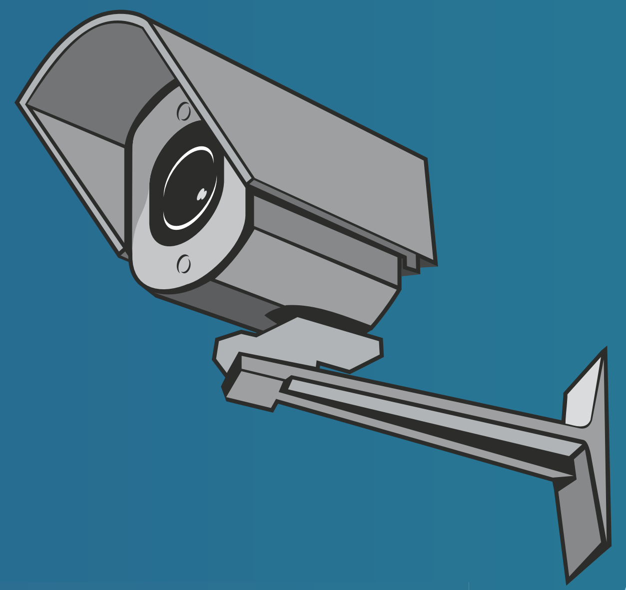
IDS (INTRUSION DETECTION SYSTEM)



- **What It Does**: IDS is like a security camera that watches the network. It detects unusual activity that might be a sign of a cyberattack and alerts the security team, but it doesn't stop the attack by itself.
- **Key Function**: Monitors and alerts the security team about potential threats.

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IPS

(INTRUSION PREVENTION SYSTEM)

- **What It Does**: IPS is like a security guard that not only watches for threats but also blocks them before they can cause harm. It's similar to IDS but with the ability to stop attacks automatically.
- **Key Function:** Blocks threats as soon as they are detected.

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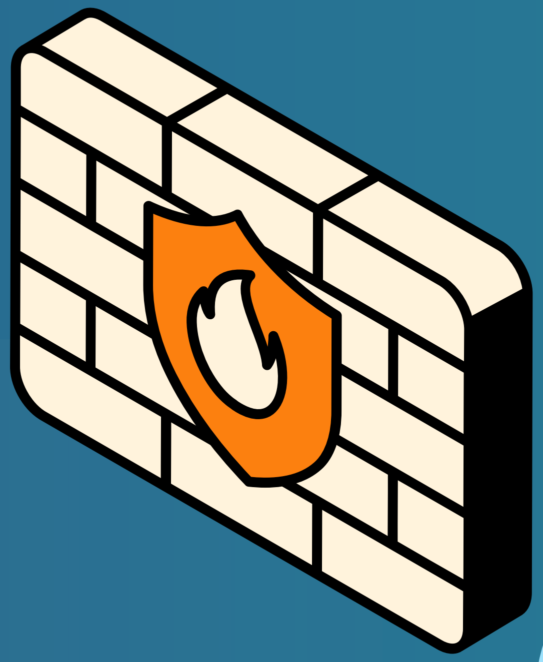
FIREWALL

- **What It Does:** A firewall is like a gatekeeper. It controls what data can enter or leave a network, blocking anything that seems dangerous, like viruses or unauthorized access attempts.
- **Key Function:** Controls and filters traffic to protect the network from unwanted access.



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EDR

(ENDPOINT DETECTION AND RESPONSE)

- **What It Does**: EDR focuses on protecting individual devices, like computers or smartphones. It watches for suspicious activity on these devices, blocks or removes threats, and alerts the security team if needed.

- **Key Function**: Protects individual devices by detecting and responding to threats.

IP ADDRESS

by Lorenzo Levano

Lorenzo Levano

HOW AN IP ADDRESS WORKS

Sending and Receiving Data

- What It Means: When you send or receive data online (like watching a video or sending a message), your IP address tells the data where to go and where it’s coming from. It’s like putting your home address on a letter so the mailman knows where to deliver it.
- Example: If you’re watching a video online, the video data is sent to your device’s IP address, making sure it arrives on your screen.

192.168.1.1

Identifying Devices

- What It Means: Just like every house has a unique address, every device connected to the internet has a unique IP address. This helps the internet know where to send information, like a web page or an email, to the right place.
- Example: If you type "google.com" into your browser, your computer uses its IP address to request the webpage, and Google sends it back to your IP address.

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TYPES OF IP ADDRESSES

Private IP Address

- What It Is: A private IP address is used inside your home or business network to identify each device individually. It's like having apartment numbers within a building; each device has its own private IP address.
- Example: Your computer might have a private IP address like "192.168.1.5" within your home network.

Public IP Address

- What It Is: A public IP address is the main address used by your home or business network when it connects to the internet. It's like the main address of an apartment building, representing everyone inside.
- Example: Your internet service provider (ISP) assigns your home router a public IP address, which the entire internet can see.

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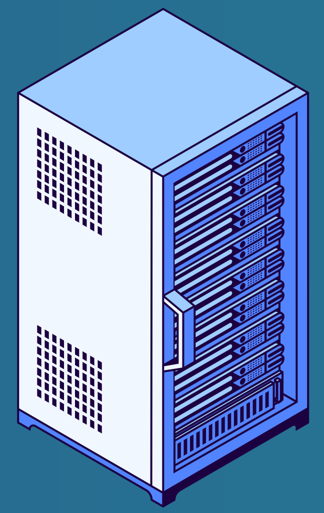
WHY
IP ADDRESSES ARE IMPORTANT



- **Routing Data**: IP addresses help route data to the correct devices, ensuring that the information you request online (like loading a website) ends up on your screen.
- **Communication**: IP addresses allow devices to communicate with each other over the internet, whether they are in the same room or on opposite sides of the world.
- **Security**: Knowing an IP address can help track where data is coming from or going to, which can be important for security reasons.

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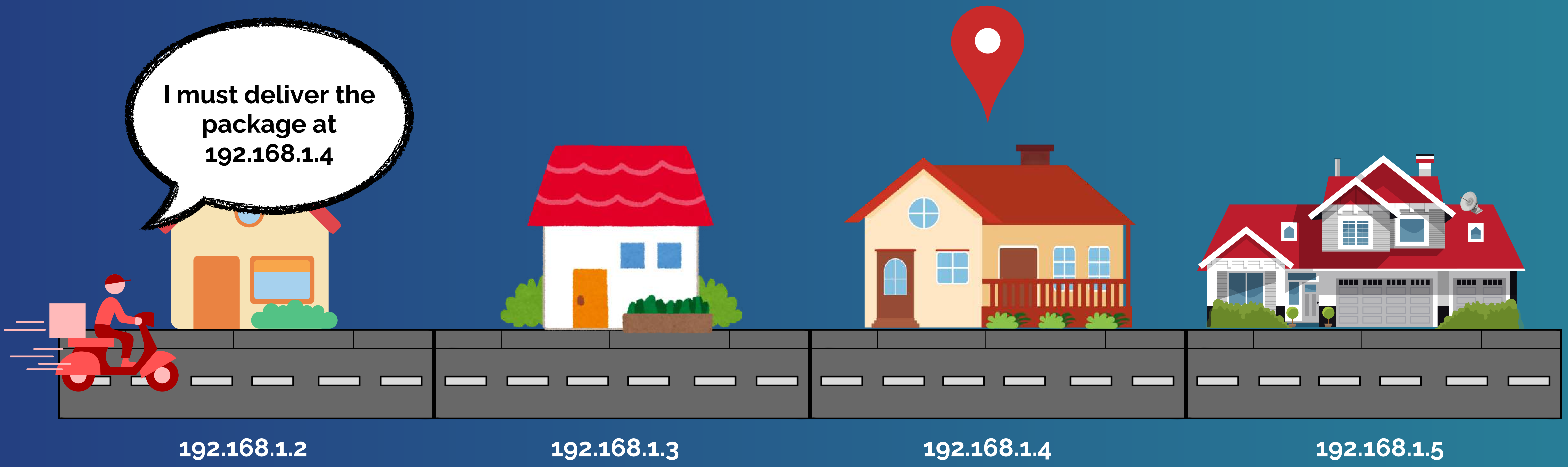
EXAMPLE OF IP ADDRESSES



WHAT IS DNS?

by Lorenzo Levano

Lorenzo Levano

HOW DNS WORKS

Turning Names into Numbers

- What It Means: When you type a website name (like www.google.com) into your browser, your computer needs to find the website's IP address (which is a set of numbers like 172.217.5.110) to load it. DNS is what helps translate the name you type into that number.
- Example: Just like you look up a friend's phone number by their name, DNS looks up the website's IP address by its name.

Making It Easy for Us

- What It Means: Instead of remembering long strings of numbers (IP addresses), you only need to remember easy names (like google.com). DNS takes care of finding the right numbers behind the scenes.
- Example: Imagine if you had to remember someone's phone number instead of just their name. DNS lets you use names while it handles the numbers.

Lorenzo Levano

WHEN YOU VISIT A WEBSITE

1 You Type the Website Name

You enter www.example.com into your browser.

2 DNS Looks Up the Address

Your computer sends a request to a DNS server (like asking for directions) to find out the IP address of www.example.com.

3 DNS Finds the IP Address

The DNS server checks its database and finds the IP address for www.example.com.

4 Your Computer Connects to the Website

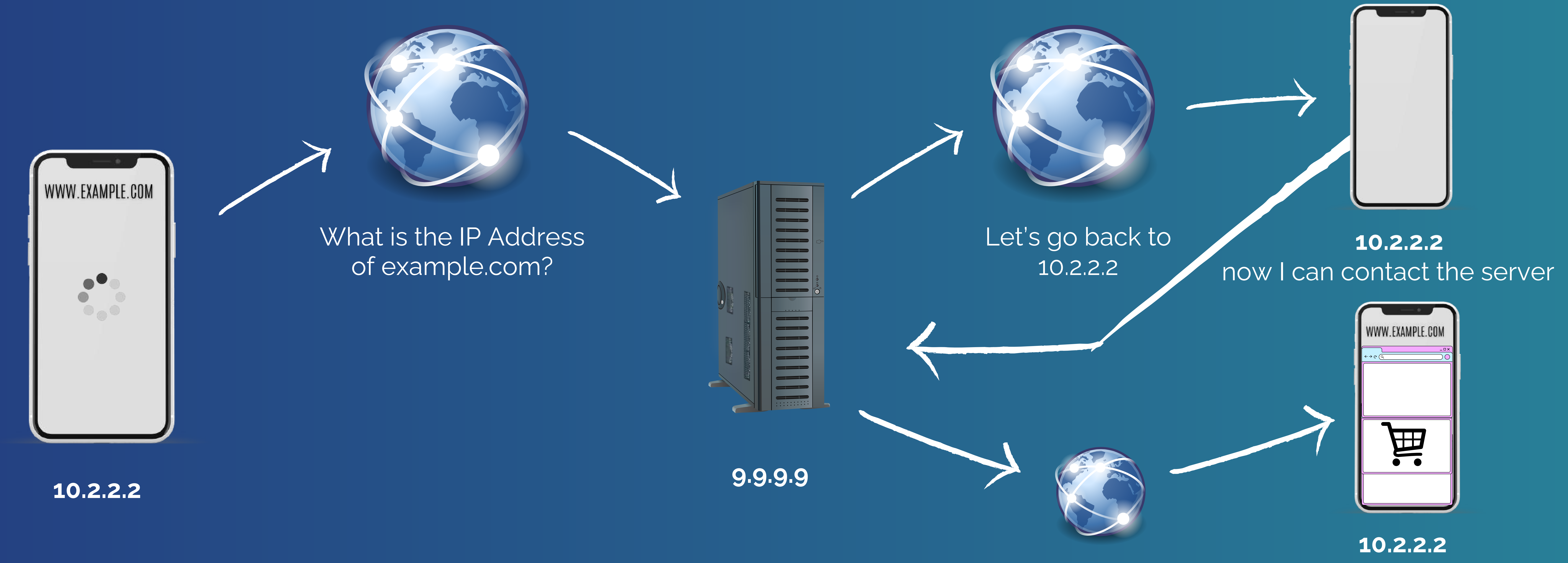
Your computer uses the IP address to connect to the website's server, and the website loads on your screen.

Lorenzo Levano

WHY
DNS IS ISMPORTANT

- **Easy to Use**: DNS lets you use simple names (like youtube.com) instead of remembering complicated numbers (IP addresses).
- **Quick Access**: DNS helps your computer find websites quickly by looking up their IP addresses in milliseconds.
- **Scalable**: DNS is designed to handle billions of requests every day, helping people all over the world access websites easily.

EXAMPLE OF DNS

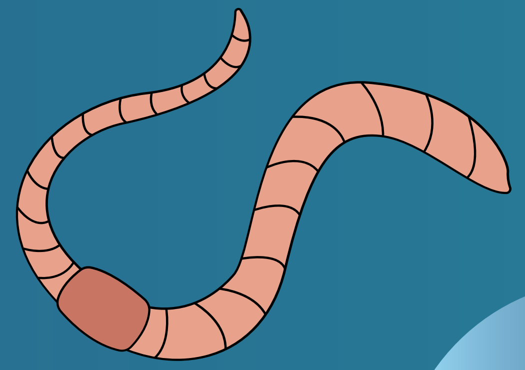


PHISHING

by Lorenzo Levano

Lorenzo Levano

HOW PHISHING WORKS (THE BAIT)



- **What It Means:** The scammer creates a fake email, text message, or website that looks like it's from a real company or person. The goal is to make you believe it's genuine.
- **Example**: You might receive an email that looks like it's from your bank, asking you to click a link to update your account information.

HOW PHISHING WORKS (THE HOOK)

- **What It Means**: The message usually contains a sense of urgency or a tempting offer to get you to act quickly without thinking. It might say your account will be locked unless you verify it right away, or that you've won a prize and need to claim it.
- **Example**: The fake email from your bank might say, "Your account has been compromised. Click here to reset your password immediately."

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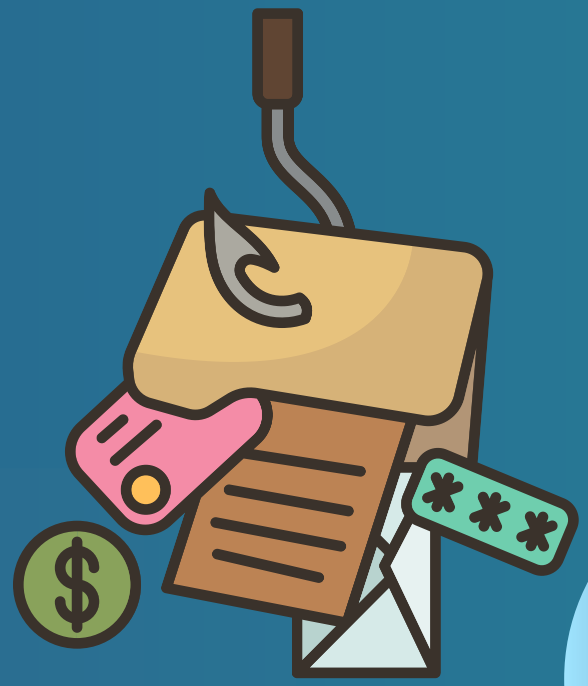
HOW
PHISHING WORKS (THE CATCH)



- **What It Means:** If you click the link and enter your information, it goes straight to the scammer. They can then use this information to steal your money, access your accounts, or even commit identity theft.
- **Example**: If you enter your bank password on the fake website, the scammer now has access to your bank account.

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WHY PHISHING IS DANGEROUS

- **Steals Personal Information**: Phishing can lead to your sensitive information being stolen, which can then be used to drain your bank account, make purchases in your name, or access other private accounts.
- **Difficult to Detect**: Phishing scams often look very convincing, making it hard to tell that they're fake. The emails and websites can look almost identical to the real thing.
- **Wide Impact**: Phishing can target anyone, from individuals to large companies, making it a widespread threat.

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HOW TO PROTECT YOURSELF

1

Check the Sender

Always check who the email or message is from. Look for small mistakes in the email address or name that might indicate it's fake.

2

Don't Click Links

Avoid clicking on links in emails or messages unless you're sure they're legitimate. Instead, go directly to the website by typing the address yourself.

3

Look for Signs

Be cautious if the message is urgent or asks for sensitive information. Real companies won't ask for personal details through email.

4

Use Security Software

Keep your computer and devices protected with up-to-date antivirus software, which can help detect and block phishing attempts.

OSINT

by Lorenzo Levano

Lorenzo Levano

COLLECTING INFORMATION

- **What It Means**: OSINT involves looking for information that is freely available to anyone. This could include things like social media profiles, news articles, public databases, and company websites.
- **Example**: If you're trying to learn more about a company, you might look at their website, social media accounts, and news articles about them.

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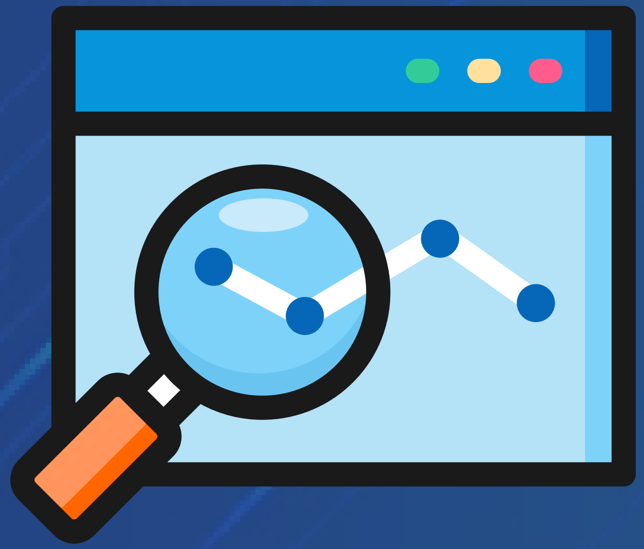
ANALYZING
DATA



- **What It Means**: Once the information is collected, it needs to be analyzed to find useful details. This might involve piecing together different bits of information to get a clearer picture.
- **Example**: You might analyze a company's social media posts to understand their latest projects or see if they've had any recent issues.

Lorenzo Levano

USING
THE INFORMATION

- **What It Means**: The gathered information can be used for various purposes, such as security assessments, business research, or even investigative work.
- **Example**: A company might use OSINT to check if their competitors are launching a new product, or a security expert might use it to identify potential threats.

Lorenzo Levano

WHY OSINT
IS USEFUL

- **Accessible Information**: OSINT uses data that is already available to the public, so there's no need for special access or permissions.
- **Cost-Effective**: Gathering information through OSINT is often cheaper than other methods because it relies on free or low-cost sources.
- **Comprehensive**: By collecting and analyzing a variety of sources, you can get a detailed and well-rounded view of the subject you're investigating.

Lorenzo Levano

HOW OSINT IS USED

Business Research

Companies use OSINT to understand market trends, monitor competitors, and find new opportunities.

Security

Security professionals use OSINT to gather information about potential threats or vulnerabilities.

Investigations

Law enforcement and investigative teams use OSINT to collect evidence and solve cases.

Lorenzo Levano

EXAMPLE OF OSINT

Imagine you're trying to learn about a new restaurant in your town:

You start by searching online for the restaurant's website and social media profiles.

You read reviews on sites like Yelp and Google Maps to see what people are saying.

You check news articles to find out if the restaurant has received any recent awards or faced any issues.

You put all this information together to get a complete picture of the restaurant.

LINUX

by Lorenzo Levano

Lorenzo Levano

WHAT IS LINUX?

Linux is an operating system, similar to Windows or macOS, that runs on computers and other devices. It's what allows your computer to work and lets you run programs and applications.

Lorenzo Levano

OPERATING SYSTEM
BASICS

- **What It Means**: An operating system is the main software that manages your computer's hardware (like the screen, keyboard, and hard drive) and provides the platform for other programs to run.
- **Example**: Just like Windows or macOS lets you use your computer, Linux does the same thing but in a different way.

Lorenzo Levano

OPEN SOURCE

- **What It Means**: Linux is "open source," which means anyone can see its code and make changes to it. This is different from Windows or macOS, which are closed source and controlled by their companies.
- **Example**: Because Linux is open source, many different versions (called "distributions" or "distros") exist, each with its own features and looks, like Ubuntu, Fedora, or Debian.

Lorenzo Levano

FREE
TO USE

- **What It Means**: Linux is free to download and use. You don't have to buy a license or pay for it, unlike some other operating systems.
- **Example**: You can download and install Linux on your computer without spending any money.

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VERSATILE
AND CUSTOMIZABLE

- **What It Means:** Linux can be customized to fit different needs, from running on supercomputers and servers to personal computers and even small devices like routers and smart TVs.

- **Example**: You can customize the look and feel of Linux to suit your preferences or needs, and use it for a wide range of tasks.

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POPULAR
IN DIFFERENT AREAS

- **What It Means**: While you might not see Linux on many personal computers, it's very popular for servers (the powerful computers that host websites and services), smartphones (Android is based on Linux), and other tech devices.

- **Example**: Most websites are hosted on servers running Linux, and many tech companies use it for their infrastructure.

WHAT IS A MALWARE?

by Lorenzo Levano

Lorenzo Levano

WHAT IS IT?

Malware stands for malicious software. It's a type of harmful computer program designed to damage, disrupt, or gain unauthorized access to your computer or personal data. Think of it as a bad actor in the digital world that tries to cause trouble on your device.

Lorenzo Levano

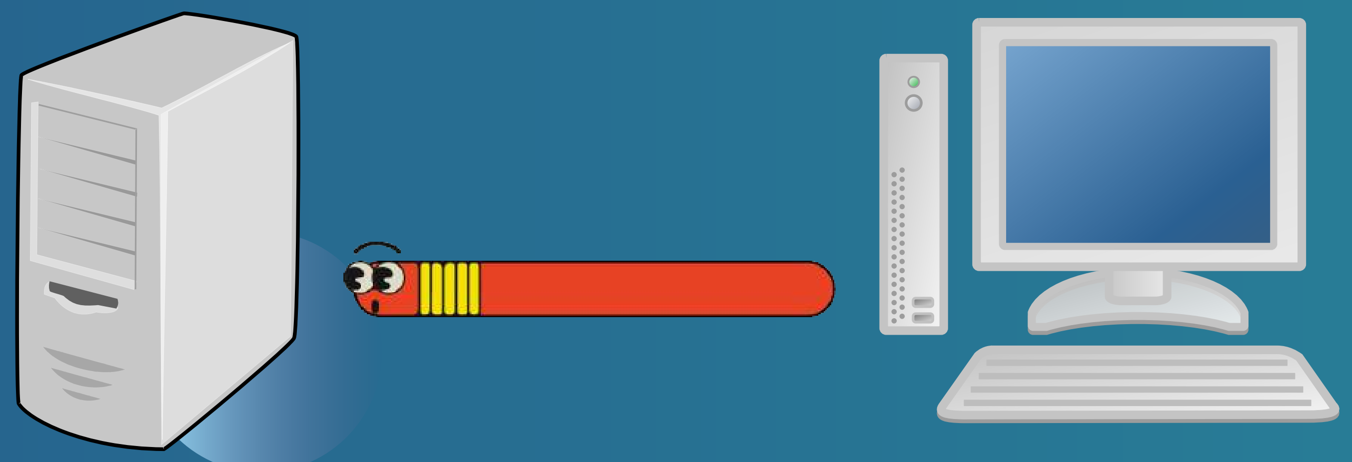
TYPES OF MALWARE

Virus

- What It Means: A virus is a type of malware that attaches itself to files or programs. It spreads when you open or share infected files.
- Example: Imagine a virus is like a sneaky germ that spreads when you touch something and then touch other things.

Lorenzo Levano

TYPES OF MALWARE

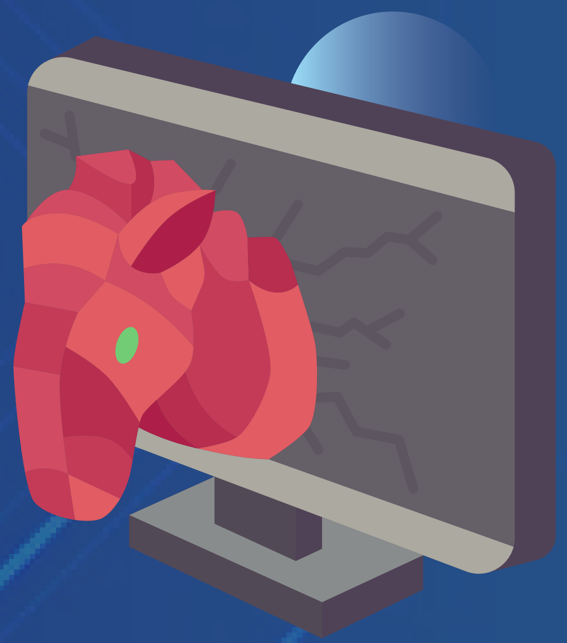


Worm

- What It Means: A worm is a self-replicating malware that spreads over networks, like the internet, without needing to attach to a file.
- Example: A worm is like a chain letter that keeps copying itself and sending out new copies automatically.

Lorenzo Levano

TYPES OF MALWARE



Trojan Horse

- What It Means: A Trojan horse pretends to be a useful or harmless program but secretly does something harmful once it's installed.
- Example: It's like a fake gift box that looks nice but contains something dangerous inside.

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TYPES OF MALWARE

Ransomware

- What It Means: Ransomware locks or encrypts your files and demands payment to unlock them.
- Example: It's like a thief who takes your belongings and demands money to return them.

Lorenzo Levano

TYPES OF MALWARE

Spyware

- What It Means: Spyware secretly monitors your activities and collects information about you without your knowledge.
- Example: It's like having a hidden camera in your home that records everything you do.

Lorenzo Levano

TYPES OF MALWARE

Adware

- What It Means: Adware displays unwanted ads and can collect information about your browsing habits to show targeted ads.
- Example: It's like a noisy ad that keeps popping up in your view, trying to get your attention.

Lorenzo Levano

HOW MALWARE
CAN AFFECT YOU

- **Slows Down Your Computer**: Malware can make your computer run very slowly by using up resources or causing errors.
- **Steals Personal Information**: It can capture your passwords, credit card numbers, or other private data.
- **Damages Files**: Malware might delete or corrupt your files, making them unusable.
- **Disrupts Operations**: It can make your computer or network stop working properly or interfere with your activities.

Lorenzo Levano

HOW TO

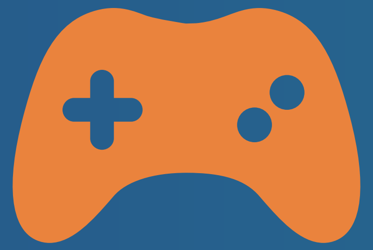
PROTECT YOURSELFR

- **Use Antivirus Software:** Install and update antivirus software to help detect and remove malware.
- **Be Cautious with Downloads**: Only download software or apps from trusted sources.
- **Keep Your System Updated**: Regularly update your operating system and software to patch security vulnerabilities.
- **Avoid Clicking on Suspicious Links**: Don't click on strange links or open email attachments from unknown senders.

Lorenzo Levano

EXAMPLE

You install the game, thinking it's just a fun app.



The game is actually malware that secretly starts stealing your personal information or messing up your files.

You notice strange behavior on your computer, like slow performance or unexpected pop-ups.